DEPI GRADUATION PROJECT

Task 5 - Implementation (Source Code & Execution)

Automated Teacher Workflow with n8n

Introduction

The Automated Teacher Workflow with n8n is an AI-enabled educational automation project designed to reduce repetitive lesson-preparation tasks. The workflow receives instructional information from a teacher, validates required fields, passes the information through AI-powered processing stages, and organizes the generated materials for teacher review.

The workflow is intended to generate a lesson outline, student activities, assessment questions, an answer key, and differentiated learning materials. It integrates n8n with Google Sheets, Google Gemini, Google Docs, and Gmail.

1. Source Code

1.1 Project Overview

The workflow is represented by an exported n8n JSON file. This export records the configured nodes, their parameters, connections, prompts, and data-processing expressions. It provides a source artifact that can be reviewed, imported, and maintained, subject to the availability of valid credentials and compatible service configurations.

1.2 Main Implementation Components

- Input processing: Receives the teacher's subject, grade level, topic, learning objective, lesson duration, and differentiation preference.
- Input validation: Uses an IF node to check whether required fields contain values before AI content generation.
- AI content generation: Uses Google Gemini nodes to generate the lesson outline, student activities, assessment questions, and answer key.
- Workflow logic: Connects processing nodes and uses conditional branches to control execution.
- Output assembly: Combines generated content into a final content field.
- Document generation: Creates a Google Docs document and inserts the generated lesson materials.
- Email notification: Uses Gmail nodes for missing-information and final-output notifications.
- Data logging: Appends teacher input records to Google Sheets.

1.3 Structured and Maintainable Code

The workflow is organized into separately named nodes, including Edit Fields, AI-Lesson outline, Students Activities, AI Assessment, Assessment Answer key, and Create a document. Each node has a defined responsibility. This modular structure makes the workflow easier to understand, troubleshoot, maintain, and extend.

The exported JSON file serves as the main source-code artifact. The workflow also uses n8n expressions to pass values between nodes and reuse teacher-provided information throughout the automation.

1.4 Coding Standards and Naming Conventions

Node names generally describe their functions, and expressions are used to retrieve values from incoming items or earlier node outputs. For example:

```
{{ $json.Subject }}
```

This expression retrieves the Subject value from the current input item. Expressions that reference prior nodes allow later AI stages to use previously generated content.

Recommended improvements include standardizing node names, field names, whitespace, output labels, and error messages. File names and documentation should follow a consistent naming convention.

1.5 Modularity and Reusability

The workflow separates educational content generation into distinct modules:

- Lesson outline generation.
- Student activities generation.
- Assessment question generation.
- Answer-key generation.
- Differentiated material generation.
- Final content assembly.
- Document creation and email notification.

This design allows individual prompts or processing stages to be improved without redesigning the entire workflow. Reusing the teacher's original input also avoids asking the teacher to enter the same information repeatedly.

1.6 Security and Error Handling

The exported workflow includes an IF node that checks required teacher inputs. The missing-information branch sends a clarification email instead of proceeding with incomplete input. Several AI nodes also have retry settings configured.

The following improvements are recommended and should be tested before the workflow is described as production-ready:

- Store API keys, OAuth tokens, and passwords in n8n credentials; do not expose secrets in source files.
- Validate required fields and ensure lesson duration is a valid positive number.
- Add clear error handling for failures involving Google Gemini, Google Sheets, Google Docs, and Gmail.
- Define a fallback path for repeated AI or integration failures.
- Avoid hardcoded recipient email addresses where configurable teacher contact information is available.
- Prevent the workflow from sending an incomplete or misleading final document.
- Review the differentiation condition to ensure that it tests the intended preference rather than comparing a field with itself.

1.7 Deliverables

The expected source-code deliverables are:

- The exported n8n workflow JSON file.
- The workflow diagram.
- Documentation describing the main nodes and their connections.
- The AI prompts used for lesson generation and assessment.
- Configuration instructions for the required integrations.
- Documentation of validation, error handling, and fallback behavior.

The workflow export should be reviewed and tested before it is treated as the final implementation.

2. Version Control & Collaboration

2.1 Version Control Repository

GitHub is recommended as the central repository for the project source code and documentation. A shared repository can hold the n8n workflow export, AI prompts, technical documentation, testing evidence, and implementation updates. The supplied files do not establish that a GitHub repository has already been created, so the repository URL should be added only after creation.

2.2 Proposed Repository Structure

```
automated-teacher-workflow/
├── README.md
├── workflow/
│   └── automated_teacher_workflow.json
├── docs/
│   ├── workflow_diagram.png
│   ├── implementation.md
│   └── configuration.md
├── prompts/
│   ├── lesson_outline.txt
│   ├── student_activities.txt
│   ├── assessment.txt
│   ├── answer_key.txt
│   └── differentiated_material.txt
├── tests/
│   └── test_cases.md
└── .gitignore
```

This structure separates workflow source code from documentation, prompts, and testing records, making the project easier to review and maintain.

2.3 Branching Strategy

A feature-branch workflow is recommended so that team members can work on separate components without directly changing the stable version.

- **main:** The reviewed and approved project version.
- **develop:** An integration branch for changes before they are promoted to main.
- **feature/workflow-updates:** Changes to n8n nodes, connections, and workflow logic.
- **feature/ai-prompts:** Improvements to AI prompts and generated educational content.
- **feature/documentation:** README files, configuration instructions, and technical documentation.
- **fix/validation-errors:** Corrections to validation and conditional logic.

Team members should commit changes to the appropriate branch and submit a pull request before merging significant changes into the stable branch.

2.4 Commit History and Documentation

Commit messages should explain changes clearly and help the team identify when specific modifications were introduced. Each commit should focus on a specific change, and important workflow updates should be accompanied by relevant documentation.

```
feat: add teacher input validation
feat: generate lesson outline using Gemini
feat: generate student activities and assessments
feat: create final lesson document in Google Docs
fix: correct differentiation condition
fix: improve missing-input notification
docs: add workflow setup instructions
test: document validation and failure scenarios
```

Pull requests should describe the purpose of the change, affected workflow nodes, tests performed, and known limitations.

2.5 Team Collaboration

The project has five team members, and responsibilities can be distributed according to the project plan:

- **Member 1 — Project Coordination:** Tracks progress, coordinates milestones, and supports the final submission.
- **Member 2 — Workflow Architecture and n8n Development:** Maintains workflow structure, node connections, and integrations.
- **Member 3 — AI Prompts and Workflow Logic:** Maintains prompts, AI transformations, and branching logic.
- **Member 4 — Testing and Quality Assurance:** Documents test cases, validates behavior, and records issues.
- **Member 5 — Documentation and Final Output:** Maintains technical documentation, user instructions, and presentation materials.

Relevant changes should be reviewed and coordinated before integration into the main workflow.

2.6 Continuous Integration and Quality Checks

Automated checks may be introduced where suitable. Recommended checks include:

- Validate that the exported workflow JSON is syntactically valid.
- Review changes to workflow nodes and connections.
- Verify that required documentation and project files are included.

- Test valid inputs, missing fields, and failure scenarios.
- Import updated workflow versions into a test environment before using them in the final environment.

Because the workflow depends on external services, successful execution also requires valid credentials and appropriate access. CI/CD pipelines are optional improvements and should only be reported as implemented if they have actually been configured and tested.

2.7 Repository Security and Backup

The repository should not contain API keys, OAuth tokens, passwords, or other sensitive credentials. The .gitignore file should exclude local secrets, temporary files, and unnecessary artifacts. Exported workflows should be reviewed for sensitive configuration details before sharing.

A reviewed workflow export should be retained at important milestones so that the team can restore a previous configuration if an update causes problems.

2.8 Deliverables

- A GitHub repository containing project source files.
- An organized repository folder structure.
- A documented branching strategy.
- A meaningful commit history.
- Pull requests or review records for significant changes.
- A README with installation and execution instructions.
- A documented final version of the n8n workflow.

These practices support collaboration, traceability, maintainability, and safer project updates.

3. Deployment & Execution- README File

README File

The Readme File Provides Project Overview and The Instructions Required to Install, configure, execute, and deploy the Automated Lesson Plan Generation System.

It is intended for developers, project evaluators, administrators, and teachers who need to understand how the system operates.

3.1 Installation Steps

Install the Required Software

There are Two installation methods, first Cloud version of n8n and the second id self Hosted like Docker Environment.

We will use n8n cloud version as the primary workflow automation platform, and create an account on, then build the workflow or import it into n8n.

3.2 System Requirements

3.2.1 Hardware:

CPU	Minimum 2 Cores (4 Cores recommended
RAM	Minimum 2-4 GB RAM (8 GB+ recommended).
Storage	5 GB storage Free.
Internet	Stable connection, High speed connection

3.2.2 Software Requirements

- Operating System Windows 10/11, Linux, or mac
- Node.js or Docker as n8n Cloud
- Google Account (Valid Gmail Account)
- Google Forms
- Google Sheets
- Google Gemini API
- Google Docs
- Web Browser

3.2.3 Software Dependencies

The system depends on:

n8n cloud account	Responsible for connecting all system components and executing the automation workflow.
Google Gemini API	As AI Engine Responsible for generating lesson outlines, student activities, assessments, answer keys, and differentiated educational content.
Google Workspace APIs	• Google Forms (Input Interface) • Google Sheets (Logging Database) • Google Docs (Output Document Builder) • Gmail (Notification & Delivery System)
Web Browser	Updated Google Chrome, Firefox or Edge

3.3 Configuration Instructions

Before executing the workflow, all external services must be configured correctly.

1. Create a Google Forms as it acts as the main input interface for teachers.
2. Create Google Sheets to Store teacher's submissions and trigger the n8n workflow with newly added rows and start the lesson generation process.
3. Google Gemini API Configuration
    - Step 1: Obtain a Gemini API Key
    - Step 2: Configure Gemini in n8n

3.4 Execution Guide

- **Node 1: Google Form (Teacher Imputes) :** The Teacher submits a Google Form containing key metadata: Subject, Grade Level, Topic, Learning Objective, Lesson Duration, and Differentiation Preference.
- **Node 2: Google Sheets Trigger:** The form submission triggers the n8n automation engine to check the connected spreadsheet for new rows add, when a new teacher request is detected, the data is passed to the next node.
- **Node 3: Data Preparation.** Edit and recreates and renames form fields for internal processing
- **Node 4: Validation Input (IF):** The system checks and ensure that all required information is available before generating the lesson plan like (The teacher's email, address is provided, the subject is specified, the grade level is selected, the lesson topic is provided, Learning objectives are available, Lesson duration is a positive number.)
    - IF Execution Conditions is Valid The workflow continues to the AI generation.
    - IF Execution Conditions are invalid the workflow moves to the clarification email stage by Triggering a Gmail notification to the teacher requesting clarification for missing fields.
- **Node 5: AI Lesson Outline:** Generate the lesson structure lesson outline.
- **Node 6: AI Student Activities:** Generate interactive educational activities aligned with the learning objectives
- **Node 7: AI Assessment:** Automatically generate A structured evaluation materials (MCQs, short answer, application questions) to evaluate student understanding aligned with the learning objectives.
- **Node 8: AI Answer Key:** Generate an accurate answer key and explanations corresponding to the assessment questions.
- **Node 9: Differentiation (IF):** Determine whether the teacher requested differentiated learning activities.
    - IF True: The workflow activates the differentiated content.
    - IF False: The workflow skips the differentiation.
- **Node 10: AI Differentiated Version:** Generate adapted activities to support students with different learning needs.
- **Node 11 – Merge (Combine Outputs):** Combine the generated components into One structured lesson plan object containing all generated sections like (Lesson Outline, Student Activities, Assessment Questions, Answer Key)
- **Node 12: Edit Fields:** The system Dynamically Prepare and organizes the lesson content into a consistent format follow a standardized structure.

- **Node 13: Google Docs Creation:** The workflow Dynamically sends the formatted content to Google Docs; the system creates a new document containing the generated the completed lesson plan.
- **Node 14: Google Sheets (Append Row):** The workflow mapping the teacher's input fields information in a logging spreadsheet, so every submission is recorded even when validation fails.
- **Node 15: Gmail Request Clarification:** The input validation stage identifies incomplete or invalid information; the system here notifies the teachers when required information is missing.
- **Node 16: Gmail Sent to Teacher:** After the Google Docs document has been created, the workflow Sends a Gmail email to the teacher containing the completed final lesson plan and direct link to the Google Doc generated successfully

Execution Guide (Running Locally)

1. Start n8n Cloud account.
2. Open the Workflow Editor
3. Import the project workflow JSON.
4. Open the imported workflow
5. Verify that all nodes are imported successfully and every node shows connected credentials (no red warnings).
6. Configure Google credentials.
7. Configure the Gemini API credentials.
8. Verify the Google Sheets connection.
9. Test the workflow by Click Execute Workflow (manual test) or toggle Active to run automatically.
10. Submit a test request using Google Forms.
11. Verify successful document creation and email delivery.

3.5 API Documentation

The system uses external APIs rather than requiring a custom backend API.

The main APIs are:

API	Purpose	Authentication
Google Sheets API	Retrieve teacher requests and maintain submission records.	OAuth 2.0
Gemini API	Generate lesson outlines, classroom activities, assessment questions, answer keys, and differentiated activities.	API Key
Google Docs API	Create and edit documents	OAuth 2.0

3.6 Executable Files & Deployment Link

Executable Files

- This system is web-based automation solution built using n8n workflow and integrated cloud services.
- The workflow is named Automated Teacher Workflow. Json file.
- RAR Folder Contain all documents and executable files

Deployment Links

1. Teacher Input Form
   - https://docs.google.com/forms/d/e/1FAIpQLSese_DKFHP6xJmFg9HCyhLqn9P6r9b_mopQ4nipkbrWmgCVCw/viewform

2. n8n Platform
   - https://emandepi.app.n8n.cloud/workflow/8w9bjCQRTa98ljvH

3. Generated Lesson Plan
   - https://docs.google.com/spreadsheets/d/12NtXNeaw6yFkQJYLQdJzBFHiLA_YR-VLuoeddvgvMRo/edit?resourcekey=&gid=840199554#gid=840199554

4. Implementation Review and Corrective Actions

A review of the supplied n8n JSON export identified several areas that should be addressed or verified. These are presented as findings and recommended corrective actions; they should not be described as completed until the changes have been implemented and tested.

Finding	Recommended Corrective Action
Differentiation condition	The current IF condition compares the differentiation preference field with itself. Change it to test the intended value or route, and verify the expected branch for each supported preference.
Answer-key prompt	The current prompt says to generate an answer key based on the assessment but does not explicitly include the assessment questions. Pass the assessment output into the prompt so that answers correspond to the generated questions.
Google Sheets trigger and append operation	The trigger watches for added rows in a worksheet to which the workflow also appends records. Review this arrangement and test for repeated executions. Consider using a separate logging worksheet or an appropriate trigger design.
Workflow diagram alignment	The supplied diagram begins with a Google Form Trigger, while the exported JSON starts with a Google Sheets Trigger. Align the diagram and documentation with the actual implementation.

The trigger and append arrangement is particularly important to review because monitoring and writing to the same worksheet can cause repeated workflow executions depending on the trigger behavior and configuration. This scenario should be tested after changes.

4.1 Verification Checklist

- Submit a complete teacher input and verify that all intended content sections are generated.
- Submit an input with a required field missing and verify that the clarification path is used.
- Test each supported differentiation preference and confirm the correct branch.
- Verify that the answer key corresponds to the actual assessment questions.
- Check that the workflow does not repeatedly trigger itself when it writes to Google Sheets.
- Confirm that the generated Google Docs document contains the expected final content.
- Verify email notification behavior and check for credential or configuration errors.

Only tests that have actually been executed should be marked as passed in the final report.

5. Presentation Summary

5.1 Source Code & Implementation

- **Objective:** Automate repetitive lesson-preparation tasks using AI and workflow automation.
- **Input Processing:** Collect teacher information and validate required fields.
- **AI Integration:** Generate lesson outlines, student activities, assessments, and answer keys using Google Gemini.
- **Workflow Logic:** Connect nodes and use conditions to control execution.
- **Output Generation:** Combine content, create a Google Docs document, and send an email notification.
- **Data Logging:** Store teacher input records in Google Sheets.
- **Code Quality:** Use modular nodes, reusable expressions, and configurable prompts.

5.2 Suggested Speaker Notes

“In Task 1, we focused on implementing the Automated Teacher Workflow using n8n. The main goal is to reduce the time teachers spend preparing lesson materials manually. The workflow collects the teacher’s information and validates the required fields. Then, Google Gemini generates the lesson outline, student activities, assessment questions, and answer key. After that, the workflow combines the generated content, creates a Google Docs document, and sends an email notification. We organized the workflow into separate nodes so that each step has a specific responsibility. This structure makes the workflow easier to understand, maintain, and improve.”

5.3 Version Control & Collaboration

- **Version Control:** Store the n8n JSON export, prompts, and documentation in a shared GitHub repository.
- **Branching Strategy:** Use separate branches for workflow development, AI prompts, documentation, and bug fixes.
- **Commit History:** Record changes with clear, meaningful commit messages.
- **Code Review:** Review important changes before merging into the main branch.
- **Team Responsibilities:** Distribute workflow development, prompt design, testing, and documentation.
- **Security:** Keep API keys, passwords, and OAuth credentials out of the repository.

Expected outcome: Better collaboration, traceable changes, easier maintenance, and safer project updates.

5.4 Suggested Speaker Notes

“In Task 2, our focus is version control and team collaboration. We plan to use GitHub as a central repository for the workflow export, AI prompts, documentation, and testing records. A clear branching strategy will help team members work on different components independently. Meaningful commit messages and code reviews will make changes easier to track and verify. We will also protect sensitive credentials and keep the project documentation updated. These practices will help us maintain the workflow and make future improvements more manageable.”

6. Conclusion

Task 1 documents the source-code structure and core implementation of the Automated Teacher Workflow, including input validation, AI-powered content generation, conditional logic, output assembly, Google Docs creation, Gmail notifications, and Google Sheets logging.

Task 2 defines a proposed version-control and collaboration process based on a shared repository, feature branches, meaningful commits, code review, security practices, and clear team responsibilities.

The technical review identifies four important areas for verification: differentiation logic, answer-key prompt context, the Google Sheets trigger/append arrangement, and consistency between the workflow diagram and the exported JSON. Corrective actions should be implemented and tested before being reported as complete.

Project reference: n8n workflow export and project documentation supplied for the Automated Teacher Workflow. GitHub and CI/CD practices in this report are recommendations unless separately verified.